

Certified Cold-Start Defect Inspection — Complete Project Document

Sep 23, 2026 · @Someone

1. Abstract

A new inspection line needs a certified defect-escape rate on day one, but it has only 5-40 real defect images to certify with; this project gives it a two-tier escape certificate built from those few real defects plus diffusion-generated synthetic ones.

Problem. Conformal risk control can certify that a detector lets at most a fraction α of defective parts escape. Its finite-sample penalty is $1/(m+1)$, where m counts labelled *defective* calibration images. With $m = 15$ it cannot certify $\alpha = 0.05$ at all, and at $\alpha = 0.10$ its only certifiable threshold is the single most conservative one, which sends nearly every part to a human.

Idea. Reduce the detector's escape event to one scalar per image, then calibrate that scalar with the exact rank-coupled transporter of Synthetic-Powered Predictive Inference (SPI, NeurIPS 2025), using defects synthesised by few-shot diffusion generators. Issue two statements, not one:

- **Tier H (hard):** a distribution-free escape cap that holds however bad the synthetic defects are. It depends only on m , the synthetic count N , α and a tolerance β , so a plant can compute it before collecting any data.
- **Tier N (near-nominal):** escape at most $\alpha + \beta + \epsilon$, where ϵ measures how far synthetic escape scores sit from real ones. This is the tight operating point, valid when the generator is good.

Correction to the earlier proposal. Re-reading SPI's theorems showed the earlier proposal overstated the guarantee. Bad synthetic data does *not* leave coverage at the nominal level; it falls back to the Tier H guardrail, which can be well below nominal. This document is built on the corrected statement.

Claim to be tested. At a fixed Tier H cap, SPERC reaches a usable review load with fewer real defects than conformal risk control, on KolektorSDD2 and Magnetic-Tile, with realised escape inside both tiers across 500 calibration resamples.

2. Problem statement

In one sentence: certify, before a line has produced enough scrap to learn from, that a frozen object detector lets through at most a chosen fraction of defective parts, and refuse when that cannot be certified.

2.1 Setting

A part image X may contain defects with ground-truth boxes $G(X)$. A frozen detector f returns boxes with confidence scores. A routing rule with threshold λ auto-accepts a part when no detection clears λ , and sends it to review otherwise. The detector is trained on data that never includes the calibration images.

2.2 Two escape events, defined once

The repository currently uses three different escape definitions (Section 13). This project fixes two, both 0/1 per image, and reports them separately.

- **Part escape (primary).** A defective part is auto-accepted. This is what reaches the customer.
- **Localised escape (secondary).** Some ground-truth defect has no matching detection (IoU at least τ , default 0.5) that clears λ . Stricter; needed for rework and root-cause.

Each reduces to one scalar per defective image:

$$\begin{aligned} c_i^{\text{part}} &= \max_{d \in f(X_i)} \text{score}(d), \quad \text{quad} \\ c_i^{\text{loc}} &= \min_{g \in G(X_i)} \max \{ \text{score}(d) : \text{IoU}(d, g) \geq \tau \} \end{aligned}$$

with the convention that an empty set gives minus infinity (the part always escapes). Then escape at threshold λ is simply:

$$L_i(\lambda) = \mathbb{1}\{c_i < \lambda\}$$

2.3 The guarantee wanted

For a target α , choose $\hat{\lambda}$ from calibration data so that, for a fresh defective part exchangeable with the calibration defects:

$$\mathbb{P}(c_{m+1} < \hat{\lambda}) \leq \alpha$$

This is a rate per *defective* part. The rate per shipped part is this times the line's defect prevalence. With score $s = -c$, it is exactly split-conformal coverage: escape is the event s above the calibrated quantile. So conformal risk control, and SPI, apply without approximation.

2.4 Why cold start breaks it

With m real defective calibration images, conformal risk control certifies threshold $c_{(k)}$ (the k -th smallest real escape score) only when $(k+1)/(m+1)$ is at most α . Two consequences follow, both already measured in this repository:

1. **Refusal floor.** No threshold is certifiable when $1/(m+1)$ exceeds α : below $m = 19$ at $\alpha = 0.05$, below $m = 9$ at $\alpha = 0.10$.
2. **Quantisation.** Certifiable escape levels are multiples of $1/(m+1)$. At $m = 15$ and $\alpha = 0.10$, only $k = 0$ qualifies, so the threshold is the lowest real score and almost every part goes to review.

KolektorSDD2 has 53 defective calibration images and Magnetic-Tile 59; a newly commissioned line has fewer. **The research problem is to certify a useful threshold from m in the range 5-40, and to say exactly what can and cannot be guaranteed at that m .**

3. Research questions and objectives

Four questions, each with a measurable answer and a pre-set way to fail.

#	Research question	Objective	Measured by	Fails if
RQ1	Can synthetic defects tighten a certified escape threshold for a detector at m of 5-40, without breaking the guarantee?	Implement SPI's exact transporter on escape scores and test it	Review load at the issued threshold vs. conformal risk control, at equal m	No tightening in the best-case test (real defects standing in for synthetic)

#	Research question	Objective	Measured by	Fails if
RQ2	What exactly is guaranteed when the generator is bad?	Turn SPI's worst-case bound into a hard escape cap for industrial use	Realised escape under deliberately misaligned synthetic data vs. the Tier H cap	Realised escape exceeds Tier H in more than the stated fraction of resamples
RQ3	Does a better generator buy certification with fewer real defects?	Compare a training-free one-shot generator with a fine-tuned few-shot one	Real defects needed (m-star) to reach a target review load at a fixed Tier H cap	No difference between generators, or alignment scores that do not predict m-star
RQ4	How many real defects must a new line collect before it can be certified?	Produce a commissioning table computable before data collection	Tier H cap as a function of m, N, alpha, beta, checked against held-out outcomes	Planned and realised caps disagree

Objectives not pursued. Beating published mAP on NEU-DET is out of scope: the claim is denominated in real labelled defects, not detection accuracy. Faithfulness-gated triage is dropped from the contribution list (Section 13 explains why).

4. Base papers

Every row below was opened and checked for authors, venue and content on 2026-09-23. Two primary base papers carry the method; the rest carry the data, the domain and the detector.

Primary base papers (method).

Paper	Venue	What this project takes from it
Bashari, Lotan, Lee, Dobriban & Romano, <i>Synthetic-Powered Predictive Inference</i>	NeurIPS 2025	The rank-coupled transporter, Theorem 3.3 (near-nominal bounds), Theorem 3.5 (worst-case guardrail), Algorithm 4 (beta selection), Corollary 3.6 (synthetic data may depend on real data)
Angelopoulos, Bates, Fisch, Lei & Schuster, <i>Conformal Risk Control</i>	ICLR 2024	The real-data-only baseline and its expected-loss guarantee for monotone bounded losses

Journal base papers — IEEE, Springer, Elsevier.

Paper	Publisher / journal	Role
Božič, Tabernik & Skočaj, <i>Mixed supervision for surface-defect detection: from weakly to fully supervised learning</i>	Elsevier, <i>Computers in Industry</i> , 2021	Introduces KolektorSDD2, the primary dataset
Huang, Qiu & Yuan, <i>Surface defect saliency of magnetic tile</i>	Springer, <i>The Visual Computer</i> 36:85-96, 2020	Introduces Magnetic-Tile, the second primary dataset
He, Song, Meng & Yan, <i>An end-to-end steel surface defect detection approach via fusing multiple hierarchical features</i>	IEEE, <i>Transactions on Instrumentation and Measurement</i> 69(4):1493-1504 (online 2019, issue 2020)	NEU-DET detection benchmark (74.8 / 82.3 mAP with ResNet34 / ResNet50)
Bergmann, Batzner, Fauser, Sattlegger & Steger, <i>The MVTec Anomaly Detection Dataset</i>	Springer, <i>International Journal of Computer Vision</i> 129:1038-1059, 2021	MVTec AD, the bridge dataset (15 categories, 5,354 images)

Paper	Publisher / journal	Role
Yuan, Li, Wang & Zhang, <i>Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems</i>	Elsevier, <i>Reliability Engineering & System Safety</i> 274:112417, 2026	Shows conformal guarantees are publishable in a top Elsevier reliability journal; time-series, not vision, no scarcity
Fan et al., <i>Few-shot mask-guided controllable anomaly image generation via inpainting diffusion for surface defect detection</i> (MaCoDiff)	Springer, <i>Journal of Intelligent Manufacturing</i> , 2026	Candidate synthetic-defect generator; no guarantee
Mi et al., <i>Joint geometry-texture decoupled diffusion for few-shot industrial anomaly synthesis</i> (JGTDiff)	Springer, <i>Journal of Intelligent Manufacturing</i> , 2026	Candidate generator, evaluated on MVTec AD; no guarantee
Angelopoulos, Bates, Candès, Jordan & Lei, <i>Learn then Test</i>	<i>Annals of Applied Statistics</i> 19(2), 2025 (IMS)	The high-probability variant already in the codebase
DSAT: a dynamic sparse attention transformer for steel surface defect detection	Springer Nature, <i>Scientific Reports</i> , 2025	NEU-DET accuracy reference: 83.14% mAP@0.5

Conference base papers.

Paper	Venue	Role
Zhao et al., <i>DETRs Beat YOLOs on Real-time Object Detection</i> (RT-DETR)	CVPR 2024 (IEEE/CVF)	Main detector; NMS-free, so calibration targets the model's own output
Xu et al., <i>Training-Free Industrial Defect Generation with Diffusion Models</i>	ICCV 2025 (IEEE/CVF)	One-shot, training-free generator — the low-cost arm of RQ3

Paper	Venue	Role
Hu et al., <i>AnomalyDiffusion: Few-Shot Anomaly Image Generation with Diffusion Model</i>	AAAI 2024	Fine-tuned few-shot generator with aligned masks, MIT-licensed code — the high-fidelity arm of RQ3
Andeol, Mossina, Mazoyer & Gerchinovitz, <i>Conformal Object Detection by Sequential Risk Control</i>	arXiv 2025	Closest detection-side conformal work

A note on venues. SPI and conformal risk control are conference papers, but NeurIPS and ICLR are the top venues of the field and are accepted as base papers by IEEE, Springer and Elsevier journals. Shen & Liu's conformal segmentation paper appeared in MDPI *Mathematics* and SH-DETR in *PLOS One*; both are cited as related work, not as base papers.

5. Literature review and research gap

Three literatures each hold one third of the problem, and none cites the others.

Thread A — guarantees for detectors. Conformal risk control (ICLR 2024) and Learn then Test (*Annals of Applied Statistics*, 2025) give finite-sample, distribution-free control of expected loss. Andeol et al. applied it to detection in 2023 with box and pixel coverage losses, then generalised it to several thresholds at once with SeqCRC (2025). Shen & Liu (2025) controlled pixel-level false discovery and miss rates for steel segmentation. Yuan et al. (*Reliability Engineering & System Safety*, 2026) controlled false-alarm rates for industrial time series. All assume the calibration pool is whatever size the dataset provides.

Thread B — guarantees under scarcity. SPI (NeurIPS 2025) is the only method built for a small real calibration set augmented with synthetic data. It is evaluated on image classification and tabular regression only. Weighted conformal prediction (Tibshirani et al., NeurIPS 2019) handles covariate shift, and this repository measured why it fails for defect-seeking acquisition: the weight ratio reached 2,103,520 at a budget of 100. Newer theory extends conformal risk control to non-monotone losses (arXiv:2604.01502, 2026), and confirms monotone losses — this project's case — keep exact control with no extra term.

Thread C — synthetic defects for cold start. Few-shot defect generation is a crowded, fast field: AnomalyDiffusion (AAAI 2024), training-free one-shot generation (ICCV 2025), MaCoDiff and JGTDiff (Springer *Journal of Intelligent Manufacturing*, 2026), and a new-product-introduction pipeline that lifted RF-DETR mAP from 78.8% to 83.3% using 3-5 reference defects (arXiv:2604.22850, 2026). VLM-based zero-shot inspection such as IAD-R1 (2025) is growing too. Every one of these reports accuracy, fidelity or diversity. None reports a risk bound.

Work	Formal guarantee	Detection	Scarce real defects	Uses synthetic defects
Conformal risk control, ICLR 2024	Yes	General	No	No
Andeol et al. 2023; SeqCRC 2025	Yes	Yes	No	No
Shen & Liu 2025	Yes	Segmentation	No	No
Yuan et al., RESS 2026	Yes	No (time series)	No	No
SPI, NeurIPS 2025	Yes	No	Yes	Yes
Non-monotone CRC 2026	Yes	General	No	No
AnomalyDiffusion; ICCV 2025 training-free; MaCoDiff; JGTDiff	No	Anomaly / segmentation	Yes	Yes
NPI defect synthesis 2026	No	Yes	Yes	Yes
IAD-R1 (VLM) 2025	No	Anomaly reasoning	Partly	No
This project (SPERC)	Yes, two-tier	Yes	Yes	Yes

The gap. No published method certifies a detector’s defect-escape rate from a small real calibration set by using synthetic defects, and none states what is still guaranteed when those synthetic defects are poor. Thread C produces the synthetic defects but never certifies. Thread B certifies with synthetic data but never on detection. Thread A certifies detection but never under scarcity.

Limit of this check. The search covered arXiv, IEEE/CVF open access, SpringerLink, ScienceDirect abstracts and two maintained survey lists. Paywalled full texts were not all readable. Re-run the intersection terms in Scopus and Web of Science before submission.

6. Proposed method: SPERC

SPERC (Synthetic-Powered Escape-Risk Certification) issues a two-tier escape certificate for a frozen detector from m real defective images and N synthetic ones. Four contributions, each tied to a research question.

C1 — Exact scalar reduction of detection escape (RQ1). Both escape events in Section 2.2 become a single score per image, so the detector problem becomes split-conformal coverage with no approximation. Missed defects are scored minus infinity and handled as permanent escapes. This makes SPI's theory apply to detection as written, rather than through a new, unproven "curve transporter".

C2 — The two-tier certificate (RQ2). SPI's theorems give two different statements. SPERC reports both, in the plant's units:

- *Tier H, hard cap.* Escape at most 1 minus SPI's Theorem 3.5 lower bound. Holds for any synthetic data, even data built from the calibration defects (Corollary 3.6). Depends only on m , N , α , β .
- *Tier N, near-nominal.* Escape at most $\alpha + \beta + \epsilon$ (Theorem 3.3), ϵ being the total-variation distance between real and synthetic order statistics. To target escape a , run SPI at $\alpha = a - \beta$.
- *Refusal.* If no β gives a Tier H cap at or below the plant's maximum tolerable escape, SPERC refuses, exactly as conformal risk control does today.

C3 — A finding that links resolution to validity (RQ2, RQ4). Tier H caps are counts over $m+1$ ranks, so they stay quantised at steps of $1/(m+1)$, like conformal risk control. Only Tier N gains the synthetic grid's $1/(N+1)$ resolution. Stated plainly: *synthetic data can sharpen the operating point when it is good, but it cannot sharpen the distribution-free worst case below the real-data grid*. This connects SPI to the quantisation bottleneck this repository already measured (Section 2.5 of OVERVIEW.md), and it is the result a reviewer from either field is least likely to have seen.

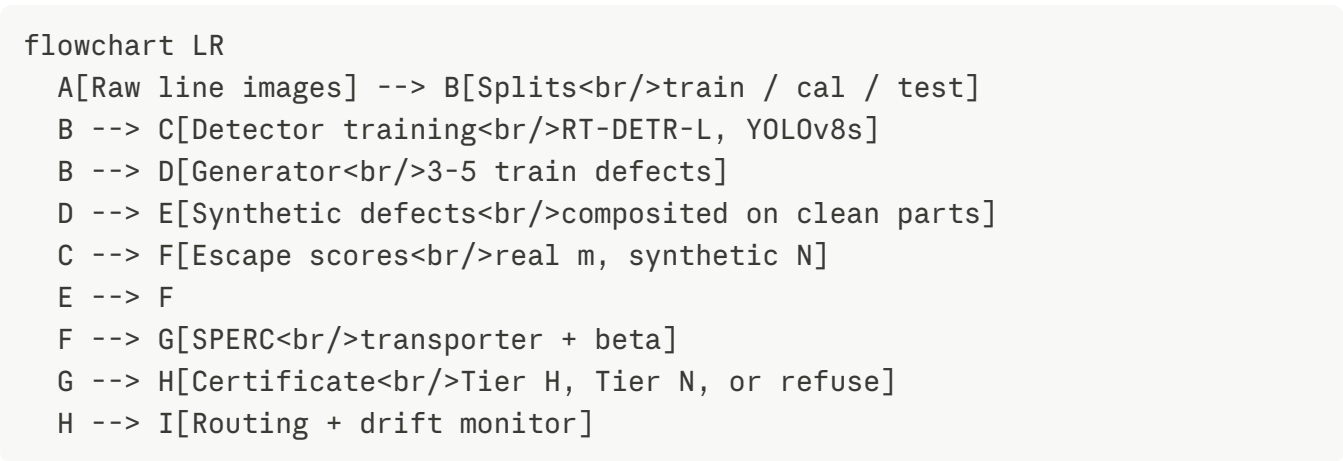
C4 — Generator quality measured in certification currency (RQ3, RQ4). Compare a training-free one-shot generator (ICCV 2025) with a fine-tuned few-shot one (AnomalyDiffusion, AAAI 2024) by the number of real defects each saves at a fixed Tier H

cap. Report an alignment diagnostic (two-sample distance between real and synthetic escape scores) and test whether it predicts that saving. Publish the commissioning table (Section 8.4) so a plant can plan m before collecting data.

What is not claimed. No new theorem. The transporter and its bounds are SPI's. The novelty is the reduction that makes SPI apply to detection, the two-tier industrial certificate, the resolution-versus-validity finding, and the first measurement of generator fidelity in real-defects-saved.

7. System architecture

Five stages, with one hard rule: the m real calibration defects never touch the detector's training.



Reading: data flows left to right; the generator and the detector draw only on the training split, and calibration scores feed SPERC alone.

Component	Input	Output	Status
Splits (src/data/splits_and_yolo.py)	COCO JSON per dataset	Disjoint train / cal / test, stratified	Exists; defect-aware allocation implemented, not yet applied
Detector (src/evaluation/train_baselines.py)	Train split	Frozen RT-DETR-L and YOLOv8s weights	Exists; 10 of 20 configurations, single seed

Component	Input	Output	Status
Generator (new: src/synth/)	3-5 reference defects from train, clean train images	N synthetic defect images with box labels	To build
Escape scoring (src/risk/spi.py, rewritten)	Detections, ground truth	c-part and c-loc per image	Localised score exists; part score to add
SPERC core (new: src/risk/spi_exact.py)	m real scores, N synthetic scores, alpha, beta	Threshold, Tier H cap, Tier N bound, or refusal	To build (Section 11)
Routing and drift (src/risk/triage.py, adaptive.py)	Threshold, live score stream	Accept / review decisions, drift alarms	Exists; reused unchanged

8. Mathematical formulation and guarantees

All formulas below are SPI's, checked against the paper's HTML on 2026-09-23 and re-derived numerically (the rank distribution sums to 1 at every m, N, r tested).

8.1 Scores

Use nonconformity $s = -c$, so a large s is a hard-to-catch defect. Real scores $s_1 \dots s_m$ come from the m calibration defects; synthetic scores $s_{\sim 1} \dots s_{\sim N}$ from generated defects scored by the same frozen detector.

8.2 The rank-coupled window

For real rank r in $1 \dots m+1$, the position of that real order statistic among the N synthetic ones follows

$$p_{\{m,N,r\}}(k) = \frac{\binom{k+r-2}{r-1} \binom{N+m-k-r+2}{m-r+1}}{\binom{N+m+1}{m+1}}, \quad \text{quad } k = 1, \dots, N+1, \quad \text{quad } F(t) = \sum_{k=1}^t p_{\{m,N,r\}}(k)$$

and the window at tolerance beta is

$$R_{r^-} = \max\{t : F(t-1) \leq \beta/2\}, \quad R_{r^+} = \min\{t : F(t) \geq 1 - \beta/2\}$$

The window depends only on m , N and β , never on score values. Real scores are transported into synthetic rank space through these windows (the transporter T in Section 3 of the earlier proposal), and the threshold is a synthetic order statistic.

8.3 What is guaranteed

Tier N (SPI Theorem 3.3). With ϵ the total-variation distance between the $(m+1)$ -th order-statistic laws of real and synthetic scores:

$$\alpha - \beta - \epsilon - \frac{1}{N+1} \leq \mathbb{P}(\text{escape}) \leq \alpha + \beta + \epsilon$$

Tier H (SPI Theorem 3.5), for any synthetic data:

$$\mathbb{P}(\text{escape}) \leq 1 - \frac{\left| \{j \in [m+1] : R_j^+ \leq \lceil (1-\alpha)(N+1) \rceil \} \right|}{m+1}$$

Both are statements about a fresh defective part, averaged over the calibration draw, as for conformal risk control. Escape here is one minus SPI's coverage, because of the reduction in Section 2.3.

8.4 Commissioning table (computed, not measured)

Tier H cap at $N = 1,000$ synthetic defects, from the formula above. The script is reproduced in Section 11. Cross-check: at $m = 15$, $\alpha = 0.05$, $\beta = 0.05$ it gives a coverage floor of 0.812, consistent with the paper's example of reaching at least 80% at that setting.

Real defects m	Conformal risk control at $\alpha = 0.10$	SPERC Tier H cap, $\alpha 0.05$, $\beta 0.05$	SPERC Tier H cap, $\alpha 0.05$, $\beta 0.20$	SPERC Tier H cap, $\alpha 0.10$, $\beta 0.05$
5	Refuses	0.333	0.167	0.333
10	Only the lowest real score is certifiable	0.182	0.182	0.273

Real defects m	Conformal risk control at $\alpha = 0.10$	SPERC Tier H cap, $\alpha 0.05$, $\beta 0.05$	SPERC Tier H cap, $\alpha 0.05$, $\beta 0.20$	SPERC Tier H cap, $\alpha 0.10$, $\beta 0.05$
15	Only the lowest real score	0.188	0.125	0.250
25	Up to the 2nd-lowest real score	0.154	0.115	0.231
40	Up to the 4th-lowest real score	0.122	0.098	0.195

How to read it. Take $m = 15$. Conformal risk control certifies 0.10 but only at a threshold that sends almost everything to review. SPERC run at $\alpha = 0.05$, $\beta = 0.05$ targets escape near 0.10 when the generator is good (Tier N: $0.05 + 0.05 + \epsilon$) and keeps escape probability at most 0.188 when it is not (Tier H; like every conformal bound, this is over the calibration draw, not per part). Raising β tightens Tier H but loosens Tier N by the same β ; the paper must report that trade-off, not hide it.

8.5 Assumptions, stated for reviewers

1. Real calibration defects are exchangeable with future defective parts (same line, same period); drift is handled by the existing monitor, not by SPERC.
2. The detector never saw the calibration defects.
3. Synthetic data may depend on anything for Tier H (Corollary 3.6). For Tier N, draw generator references from the training split, so ϵ has its plain meaning.
4. Ties among scores are broken at random, including the minus-infinity scores of missed defects.

9. Datasets, splits and the cold-start protocol

Keep all five, give them different jobs, and add MVTec AD as a bridge. Five datasets is fine for a journal paper, provided the headline rests on the two with clean parts and the other three are framed as validity checks.

Dataset	Source (verified)	Content	Role	Why
KolektorSDD2	Elsevier <i>Computers in Industry</i> 2021	3,335 images: 356 defective, 2,979 clean; 1 class; CC BY-NC-SA 4.0	Primary	89% clean, so review load is measurable; 53 defective calibration images already
Magnetic-Tile	Springer <i>The Visual Computer</i> 2020	5 defect classes plus defect-free (952 defect-free per the repository converter)	Primary replication	Second line with clean parts; 59 defective calibration images
NEU-DET	IEEE <i>TIM</i> 2020	1,800 images, 6 classes, boxes	Validity check	Every image defective: escape control is testable, review load is not
GC10-DET	MDPI <i>Sensors</i> 20(6):1562, 2020	3,570 images, 10 classes, boxes	Validity check	Wide strips, dense defects; stresses the localised-escape score
PCB (PKU)	arXiv:1901.08204, 2019	1,386 images, 6 defect types, synthetically inserted defects	Validity check	Tiny defects; note in the paper that its defects are artificial

Dataset	Source (verified)	Content	Role	Why
MVTec AD (new)	Springer <i>IJCV</i> 2021	15 categories, 5,354 images; train defect-free only; test 1,258 defective, 467 clean; pixel masks	Bridge	SPI's native setting (one anomaly score per image); lets the transporter be validated before the detection step; the generators in Section 4 report on it

Repository totals across the first five: 9,468 images, 11,543 boxes, 28 classes.

9.1 Splits

Train / cal / test, with cal never seen by training or model selection (already enforced in `src/data/splits_and_yolo.py`). Apply the defect-aware allocation that already exists (defective 45/30/25, clean 60/15/25), which roughly halves every alpha floor.

9.2 Two cold-start protocols

- **P1, calibration cold start.** Detector trained on the full train split. Subsample the real defective calibration pool to m in $\{5, 10, 15, 25, 40, \text{full}\}$, 500 random draws per m . Isolates the certification question.
- **P2, full cold start (the realistic case).** Detector trained on $k = 5$ real defects plus synthetic defects plus all clean images, as in the new-product-introduction setting; calibration on m disjoint real defects. Run on KolektorSDD2 and Magnetic-Tile only.

P1 gives the clean scientific result; P2 is what a reviewer from industry will ask for. Run P1 first.

10. Implementation on Windows with an RTX 5060

Everything in this plan fits an 8 GB RTX 5060 on Windows; the one step that needs care is generator training, and the new-product-introduction paper reports it peaking under 6 GB.

10.1 Hardware facts (verified)

Item	RTX 5060 (desktop)
Architecture / chip	Blackwell, GB206; compute target sm_120
CUDA cores	3,840
Memory	8 GB GDDR7, 128-bit, 448 GB/s
Bus / power	PCIe 5.0 x8, 145 W
Released	19 May 2025

Sources: [GeForce RTX 50 series](#); sm_120 naming per the [PyTorch Blackwell support issue](#).

10.2 Software stack

1. **Driver:** a current NVIDIA Game Ready or Studio driver that lists RTX 50 series support.
2. **Python 3.11 venv** named `vqaenv` at the repository root (the README already assumes `./vqaenv/Scripts/python.exe`).
3. **PyTorch with CUDA 12.8 or newer wheels.** Blackwell support and CUDA 12.8 wheels arrived in [PyTorch 2.7](#); the official selector lists a Windows cu128 build:

```
pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu128
```

4. **Check before any training** — if `sm_120` is missing, every later step will fail or silently run on CPU:

```
./vqaenv/Scripts/python.exe -c "import torch; print(torch.__version__,
torch.cuda.is_available(), torch.cuda.get_device_name(0),
torch.cuda.get_arch_list())"
```

5. **Pinned extras:** `ultralytics` (missing from requirements.txt today — add it), `diffusers`, `transformers`, `accelerate`, `scipy`, `pycocotools`, `opencv-python`.

10.3 Settings per task (starting points)

These are starting points, not measurements. Confirm each with a one-epoch smoke test and `nvidia-smi` before a long run.

Task	Setting on 8 GB
YOLOv8s training	Repository's reduced profile: batch 8 at imsz 256-384, batch 4 at 512-640, batch 2 at 800; AMP on
RT-DETR-L training	Start at half the YOLOv8s batch for the same imsz; RT-DETR uses deformable attention and is heavier
Prediction caching	batch 1, <code>half=True</code> ; memory is not the constraint
Fine-tuned generator (AnomalyDiffusion-style)	fp16 mixed precision, batch 1, gradient checkpointing; the NPI paper reports under 6 GB peak and about 16 h of embedding learning on an RTX 5080, so expect longer on a 5060
Training-free generator (ICCV 2025)	Inference only, fp16; about 15 s per image was reported on an RTX 5080 for the NPI pipeline, a useful order of magnitude
SPERC calibration	CPU only; milliseconds per run, so 500 resamples x 6 values of m is trivial

10.4 Windows-specific rules

- **Dataloader workers = 0.** Already set in `train_baselines.py`; Windows spawns worker processes by re-importing the script.
- **Guard every entry script** with `if __name__ == "__main__":` , for the same reason.
- **Prefer PyTorch's built-in scaled-dot-product attention** over third-party attention kernels, which can lag a new GPU architecture.
- **Keep data and runs outside OneDrive-synced folders** and enable Windows long-path support; COCO and YOLO trees exceed 260 characters quickly.
- **One GPU job at a time.** The prior 10-of-20 gap came from runs killed when another project took the GPU; use the pipeline's existing `--wait-for-gpu` flag.
- **AnomalyDiffusion's instructions target Ubuntu.** If it fails on native Windows, run only that step under WSL2 with the same GPU; everything else stays native.

11. Codebase map

Most of the pipeline exists and is reusable; the SPI module does not implement SPI and must be replaced before any SPERC result means anything.

11.1 What exists (read from the source on 2026-09-23)

Module	What it does	Keep / fix
src/data/prepare_datasets.py	VOC and mask converters for all five datasets to COCO	Keep
src/data/splits_and_yolo.py	Train / cal / test with dedicated calibration block; YOLO export	Keep; apply defect-aware fractions
src/data/build_cce_dataset.py, photometric.py	CCE encoding (measured neutral)	Keep as ablation only
src/evaluation/train_baselines.py	Per-dataset RT-DETR-L / YOLOv8s training, workers=0, per-dataset imgsz	Keep
src/risk/conformal.py	CRC, Learn then Test (Hoeffding-Bentkus), Mondrian, IoU-robust CRC	Keep; note its escape loss is a <i>fraction</i> of missed boxes
src/risk/adaptive.py	Online threshold adaptation and KS drift monitor	Keep
src/risk/triage.py	Accept / review / reject policy under LTT	Keep; drop the faithfulness gate from claims
src/risk/acquisition.py, stratified.py	Defect-seeking acquisition and its refuted corrections	Keep as baselines
src/risk/spi.py	Claims to implement SPI	Replace (below)

Module	What it does	Keep / fix
src/xai/, src/agentic/	Occlusion saliency, closed-loop agent, Ollama narrator	Keep; out of the paper's core claim
src/evaluation/rigor.py, seed_analysis.py	Wilson intervals, bootstrap stability, paired seed tests	Keep; use for all reporting

11.2 Two problems found in the code

Problem 1 — src/risk/spi.py is not SPI. Its `_rank_windows` uses a normal approximation to rank fluctuation instead of SPI's exact hypergeometric window, and `spi_threshold` never calls it. Instead it maps each real score to its nearest synthetic value from below and keeps conformal risk control's budget over the m real points. That threshold can only equal or undercut the plain CRC threshold, which explains why the earlier "Attempt B" was valid but never tighter.

Problem 2 — three escape definitions. `conformal.py` controls the expected *fraction* of missed boxes; `spi.py` controls a 0/1 "any box missed" event; the triage policy and `rigor.max_detection_score` use a 0/1 part-level event from the maximum score. They are different risks with different numbers. Section 2.2 fixes the paper on the two 0/1 events. A minor, safe issue: `crc_threshold` returns the k -th order statistic where the $(k+1)$ -th would still be certified, so it is conservative by one step.

11.3 What must be built

New file	Contents
src/risk/spi_exact.py	Exact windows (below), transporter, threshold, Theorem 3.5 cap, Algorithm 4 beta search
tests/test_spi_exact.py	Reproduce SPI's coverage on synthetic Beta scores; adversarial synthetic data must stay under Tier H
src/synth/	Wrappers for the ICCV 2025 training-free generator and AnomalyDiffusion; compositing onto clean parts; box labels from generated masks

New file	Contents
scripts/run_coldstart_sweep.py	P1 and P2 sweeps: m in $\{5, 10, 15, 25, 40, \text{full}\}$, 500 draws, both escape events

The exact window and the Tier H cap, as used to compute the table in Section 8.4:

```
import numpy as np
from scipy.special import gammaln

def _logC(n, k):
    return gammaln(n + 1) - gammaln(k + 1) - gammaln(n - k + 1)

def rank_pmf(m, N, r):
    k = np.arange(1, N + 2)
    return np.exp(_logC(k + r - 2, r - 1) + _logC(N + m - k - r + 2, m - r + 1)
                  - _logC(N + m + 1, m + 1))

def windows(m, N, beta):
    lo, hi = [], []
    for r in range(1, m + 2):
        F = np.concatenate([[0.0], np.cumsum(rank_pmf(m, N, r))]) # F[t], t = 0..N+1
        t = np.arange(1, N + 2)
        lo.append(t[F[t - 1] <= beta / 2].max())
        hi.append(t[F[t] >= 1 - beta / 2].min())
    return np.array(lo), np.array(hi)

def tier_h_cap(m, N, alpha, beta):
    _, hi = windows(m, N, beta)
    q = int(np.ceil((1 - alpha) * (N + 1)))
    return 1.0 - (hi <= q).sum() / (m + 1)
```

12. Experimental protocol

Seven stages, cheapest first; each has a stop rule, so a dead idea dies before it costs GPU days.

Stage	Experiment	GPU	Stop rule
0	<code>test_spi_exact.py</code> : SPI on synthetic Beta scores; adversarial synthetic data	None	Realised coverage outside SPI's Theorem 3.3 / 3.5 bounds = implementation bug; fix before anything else
1	Best case: held-out real GC10 defects play the synthetic role (m in {5, 10, 15, 25})	None (cached scores)	No review-load gain over CRC at equal Tier H cap = stop, write the fallback paper
2	MVTec AD bridge: image-level anomaly scores	Low	Transporter must behave as SPI reports on its own kind of score
3	Retrain with defect-aware splits, 3 seeds, primary datasets first	High (overnight jobs)	Needed regardless of SPERC's outcome
4	P1 sweep on KolektorSDD2 and Magnetic-Tile with both generators	Medium (generation)	This is the headline figure
5	Misalignment stress: another dataset's defects, an under-trained generator checkpoint, uniform noise	None	Realised escape above Tier H beyond sampling error = the claim fails
6	P2 full cold start on the two primary datasets	High	Reported as the deployment-realistic result, whatever it shows

12.1 Baselines (all but one already in the repository)

1. **CRC, real only** — `src/risk/conformal.py`.
2. **Naive stacking** — real and synthetic pooled into one CRC set; already measured invalid (0.12-0.16 escape against 0.10), kept to show why ranks matter.
3. **The current `spi.py`** — to document that it never beats CRC.
4. **Defect-seeking acquisition without correction** — `src/risk/acquisition.py`.
5. **Oracle** — CRC on the full real calibration pool, the ceiling.

6. **SPERC** — new, with beta in {0.05, 0.10, 0.20} and both generators.

12.2 Metrics

Metric	Definition	Role
Realised escape	Held-out escape rate at the issued threshold, averaged over 500 calibration draws, Wilson interval	Validity: must sit inside Tier N (aligned) and under Tier H (always)
Issuance rate	Share of draws where a threshold is issued rather than refused	Usability
Review load	Share of held-out parts sent to review at the issued threshold (primary datasets only)	The quantity a plant pays for
m-star	Smallest m at which issuance is at least 80% and review load at most a set target, at a fixed Tier H cap	The headline number, in real defects
Alignment	Two-sample distance between real and synthetic escape scores (KS statistic)	Tests whether generator quality predicts m-star (RQ3)
Detector accuracy	mAP@0.5 and mAP@0.5:0.95, mean and spread over 3 seeds	Context only; not the claim

12.3 Reporting rules

- Every number with its seed count and resample count.
- Both escape events reported, never mixed.
- Refusals reported as refusals, never dropped from averages (the prior "issued-only" selection bias is documented in OVERVIEW.md).

13. Current status and known issues

The repository has a validated core and an honest record of negative results, but eleven documentation and code issues must be fixed before a reviewer sees it.

13.1 Already established (from OVERVIEW.md and the experiment records)

Result	Evidence in the repository
Escape scalar reduction holds	CRC on the reduced score: 0.032 / 0.082 / 0.179 escape against alpha 0.05 / 0.10 / 0.20, 600 trials
CRC leaves budget unused at small m	m = 25: refuses at 0.05; uses 40% of budget at 0.10 and 77% at 0.20
Alpha floors are set by defective counts	KolektorSDD2 0.0185 (53 defective), Magnetic-Tile 0.0167 (59)
Defect-seeking acquisition buys certificates	KolektorSDD2, budget 100: 0% issuance at random vs. about 83% guided, realised 0.1011 against 0.10
Three standard bias corrections fail for one reason	Inverse-propensity (0% issuance), stratified (6.0% best valid), deflated alpha (cliff to 0%); operating grid too coarse at small m
Learned nonconformity score does not help	KolektorSDD2 auto-accept falls from 88.1% to 0.7%
Drift controller	Holds 0.1007 after a regime shift where a static threshold reaches 0.9618

13.2 Issues to fix

- ☐ Replace src/risk/spi.py with the exact SPI implementation (Section 11.2, Problem 1).
- ☐ Unify escape definitions to the two 0/1 events (Problem 2).
- ☐ README lists faithfulness-gated triage (C2) as a contribution; OVERVIEW and RESEARCH_PLAN call it vacuous. Remove it from the claims.
- ☐ Reconcile "12/12" (README), "20/20" (OVERVIEW) and "15/15" (RESEARCH_PLAN) validated operating points into one recomputed number.
- ☐ Reconcile detector-run counts: "10 of 20" vs. six named missing runs vs. "12/12 configs".
- ☐ OVERVIEW states 10 commits; git shows 33. It states 4 notebooks; the repository has 2.
- ☐ README says test_conformal passes 9/9; the file has 10 tests. README omits test_rigor (3 tests).

- ☐ requirements.txt lacks `ultralytics` and lists the removed Deformable-DETR stack.
- ☐ configs/config.yaml describes the removed Kaggle / Deformable-DETR / Swin pipeline; delete or rewrite it.
- ☐ README's layout lists `src/xai/rollout.py`, which does not exist.
- ☐ SH-DETR is quoted as 0.831 mAP; the paper reports 83.03%.

Every tracked file also shows as modified in git purely from line-ending changes; add a `.gitattributes` with `* text=auto` so real changes stay visible.

14. Risks, limitations and the fallback paper

The largest risk is that Tier H caps are too loose to be useful at small m ; the plan reports that honestly rather than hiding it, and there is a publishable paper either way.

Risk	Likelihood	Effect	Response
Tier H caps too loose to matter (e.g. 0.188 at $m = 15$)	Medium	Weakens the hard-guarantee story	Position Tier H as a planning floor and Tier N as the operating point; C3 turns the looseness itself into a result
Generator escape scores misaligned with real ones	Medium	Tier N no tighter than CRC	Stage 1 detects this cheaply; RQ3 measures it rather than assuming it away
"This is SPI applied to detection"	High	Reviewer rejection at a top ML venue	Claim the reduction, the two-tier certificate, C3 and C4 explicitly; target applied venues (Section 15)
Exchangeability fails on a real line	Medium	Both tiers invalid	Keep the drift monitor; state the assumption; do not claim per-part guarantees

Risk	Likelihood	Effect	Response
Detector mAP below state of the art (0.733 vs. 83.14% DSAT on NEU-DET)	Certain	Reviewer comment	The claim is in real defects, not mAP; report the gap
GPU contention kills long runs again	Medium	Missing configurations	One job at a time; resumable pipeline; primary datasets first
Paywalled prior work overlaps	Low-medium	Novelty claim weakens	Scopus and Web of Science search before submission

Limitations to state in the paper. Guarantees are marginal, over calibration draws, not per part. Drift in the experiments is synthetic. PCB defects are artificially inserted. Tier N depends on an unobservable distance epsilon; the KS diagnostic estimates it but does not certify it.

Fallback paper, if Stage 1 fails. "Where conformal escape control breaks at cold start, and why corrections fail": the scalar reduction, the quantisation analysis, the defect-seeking acquisition results, the three refuted corrections, and the Stage 1 negative result for synthetic transport. Every piece is already measured except the last, which is cheap.

15. Timeline and publication plan

About 12 weeks of work on one RTX 5060, with the go / no-go at week 2, then a journal submission.

Weeks	Work	Exit condition
1	Fix the 11 issues in Section 13.2; write spi_exact.py and its tests (Stage 0)	Tests reproduce SPI's bounds
2	Stage 1 best-case test on cached GC10 scores	Go / no-go. No gain = switch to the fallback paper
2-3	Stage 2 on MVTec AD	Transporter behaves as SPI reports
3-6	Stage 3: 3-seed retrains with defect-aware splits, primary datasets first, then the rest	All configurations complete, no interrupted runs recorded

Weeks	Work	Exit condition
5-8	Generators: training-free first (days), AnomalyDiffusion second (longer; WSL2 if needed)	N = 1,000 synthetic defects per primary dataset, visually audited
8-9	Stage 4 sweep and Stage 5 stress tests	Headline figure and validity figure
9-10	Stage 6 full cold start	Deployment-realistic result
10-12	Writing, Scopus / Web of Science novelty check, internal review	Submission

15.1 Where to submit

Each venue below has published directly adjacent work, checked in this survey.

Venue	Publisher	Why it fits	Evidence it publishes this kind of work
<i>Computers in Industry</i>	Elsevier	Industrial vision with a deployable system	Published KolektorSDD2 (2021)
<i>Journal of Intelligent Manufacturing</i>	Springer	Few-shot defect synthesis plus manufacturing decisions	Published MaCoDiff and JGTDiff (2026)
<i>Reliability Engineering & System Safety</i>	Elsevier	Guarantees as the headline	Published conformal anomaly detection for industrial systems (2026)
<i>IEEE Transactions on Instrumentation and Measurement</i>	IEEE	Inspection measurement and defect detection	Published the NEU-DET detection paper (2020)
COPA symposium (PMLR proceedings)	Conformal community	Short methods paper on C3, the resolution-versus-validity finding	Published Andeol et al.'s conformal object detection (COPA 2023, PMLR vol. 204)

Recommended order. Full paper to *Computers in Industry* or *Journal of Intelligent Manufacturing*; a short C3-focused paper to COPA in parallel, if the journal's policy on overlapping submissions allows it. A first-tier ML main track (NeurIPS, ICML, ICLR) is not recommended: there is no new theorem.

15.2 What must be true before submitting

1. Stage 1 passed, or the fallback paper is the one being written.
2. All Section 13.2 items closed; one number per claim across README, OVERVIEW and the paper.
3. Three seeds on both primary datasets.
4. Novelty re-checked in Scopus and Web of Science in the submission month; five of the papers cited here are from 2026.

16. References and sources checked

Every page below was opened on 2026-09-23. Numbers in this document come from these pages or from the repository itself.

Method and theory

1. [Bashari et al., Synthetic-Powered Predictive Inference, NeurIPS 2025 \(HTML, theorems\) · ar5iv render \(transporter\) · code](#)
2. [Angelopoulos et al., Conformal Risk Control, ICLR 2024](#)
3. [Angelopoulos et al., Learn then Test, Annals of Applied Statistics 19\(2\), 2025](#)
4. [Conformal Risk Control under Non-Monotone Losses, arXiv:2604.01502](#)
5. [Xu, Guo & Wei, Selective Conformal Risk Control, arXiv:2512.12844](#)

Conformal methods in detection and industry

6. [Andeol et al., Confident Object Detection via Conformal Prediction and Conformal Risk Control: an Application to Railway Signaling, COPA 2023, PMLR 204](#)
7. [Andeol et al., Conformal Object Detection by Sequential Risk Control, arXiv:2505.24038](#)
8. [Shen & Liu, Conformal Segmentation in Industrial Surface Defect Detection, arXiv:2504.17721](#)
9. [Yuan et al., Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems, RESS 274:112417, 2026](#)

Detectors and accuracy references

10. [Zhao et al., DETRs Beat YOLOs on Real-time Object Detection, CVPR 2024](#) · [Ultralytics RT-DETR docs](#)

11. [DSAT, Scientific Reports 2025](#)

12. [SH-DETR, PLOS One 2025](#)

Synthetic defect generation

13. [Hu et al., AnomalyDiffusion, AAAI 2024](#) · [code](#)

14. [Xu et al., Training-Free Industrial Defect Generation with Diffusion Models, ICCV 2025](#)

15. [Fan et al., MaCoDiff, Journal of Intelligent Manufacturing 2026](#)

16. [Mi et al., JGTDiff, Journal of Intelligent Manufacturing 2026](#)

17. [Güğöl, Levi & Acar, Few-Shot Diffusion-Based Defect Synthesis for New Product Introduction, arXiv:2604.22850](#)

18. [IAD-R1, arXiv:2508.09178](#)

Datasets

19. [KolektorSDD2 \(Božič et al., Computers in Industry 2021\)](#)

20. [Magnetic tile \(Huang et al., The Visual Computer 2020\)](#)

21. [NEU-DET detection \(He et al., IEEE TIM 2020\)](#)

22. [GC10-DET \(Lv et al., Sensors 2020\)](#)

23. [PKU PCB defects \(arXiv:1901.08204\)](#)

24. [MVTec AD \(Bergmann et al., IJCV 2021\)](#)

Hardware and toolchain

25. [GeForce RTX 50 series specifications](#)

26. [PyTorch 2.7 release \(Blackwell, CUDA 12.8\)](#) · [PyTorch install selector](#) · [sm_120 support issue](#)

Not independently opened: Tibshirani et al., Conformal Prediction Under Covariate Shift (NeurIPS 2019) is cited from the repository's own experiment records; confirm the reference before submission.
